

Is the Visual Analogue Scale an Interval Scale?

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Is the Visual Analogue Scale an Interval Scale?

Stage 1: what the published literature can actually support

Date: 10 September 2026 **Status:** search and screening complete; the analysis is blocked by data supply, not by method.

Summary

The plan was the pain analogue of the PISA score interval plot: estimate latent pain from a multi item pain instrument, regress the displayed VAS on it with a monotone spline, and show how much latent pain a 10 point displayed interval actually spans at each point on the scale.

The method works. It was validated against simulated data where the display's nonlinearity is known by construction, and it recovers the true curve.

The shared data cannot carry it. Of **4,567** open access papers that use a VAS for pain alongside a named multi item pain instrument, **493** carry a data availability statement, and **two** share the item level responses an item response model requires. Neither of those two can carry the analysis. The rest share subscale and instrument totals, which cannot be modeled.

This is a real finding about the research record rather than a failure of the search. It is also the single most useful thing to know before committing to this project: the bottleneck is not finding papers, it is that pain researchers deposit scores, not responses.

What was done

SEARCH

Europe PMC, open access with full text in EPMC, 2012 to 2026. The query required a visual analogue scale term, the word pain, and a named multi item pain instrument, all in the abstract:

Brief Pain Inventory, McGill Pain Questionnaire, WOMAC, KOOS, PROMIS, SF-MPQ, numeric rating scale, numerical rating scale, or pain intensity

Four strata were run and de-duplicated: a broad stratum taken to exhaustion (4,581 hits), a supplementary material stratum, a body text stratum matching repository names, and a journal policy stratum covering PLOS ONE, Scientific Reports, BMJ Open and PeerJ. The journal policy stratum returned **zero** records not already captured, which is reassuring about coverage: the journals with mandatory data policies were already fully inside the net.

4,570 unique PMC records.

DATA SHARING DETECTION

Every record's full text XML was downloaded and passed through

```
rttransparency::rt_data_code_pmc()
```

. 4,567 were processed successfully.

	n	% of screened
Papers screened	4,567	100%
Data availability statement detected	493	10.8%
Code sharing detected	36	0.8%
With a repository link (figshare, OSF, Zenodo, Dryad)	41	0.9%

Detection has risen steadily, from zero in 2012 and 2013 to 16.5% in 2026.

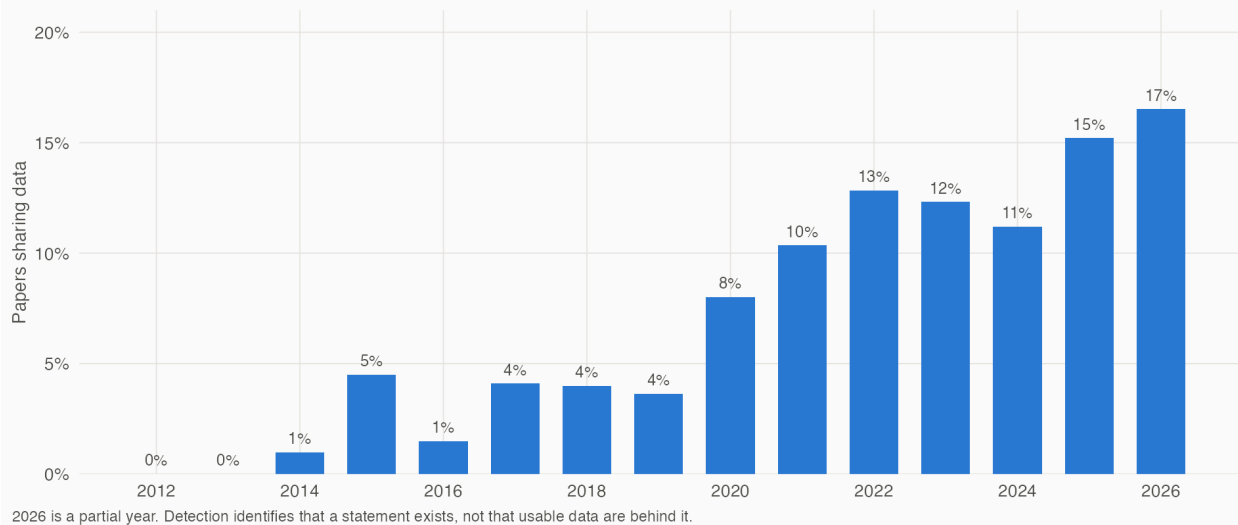
What those 493 statements actually say matters more than the count. The detector reports that an availability statement exists, not that usable data sit behind it:

Statement type	n	% of the 493
In the article or its supplement	324	65.7%
Other wording	130	26.4%
Names a repository or accession	39	7.9%

Two thirds point at the article itself, which is exactly where screening then found subscale totals rather than item responses. Any headline rate quoted from this corpus should say “availability statement detected”, not “data shared”.

Detected data availability statements rose from nothing to one paper in six

Share of open access pain papers carrying a detected data availability statement or repository deposit, by publication year. 4,567 papers, detection by transparency. Most statements point at the article itself, not at a repository: only 39 of 493 named a repository or accession.



RETRIEVAL AND SCREENING

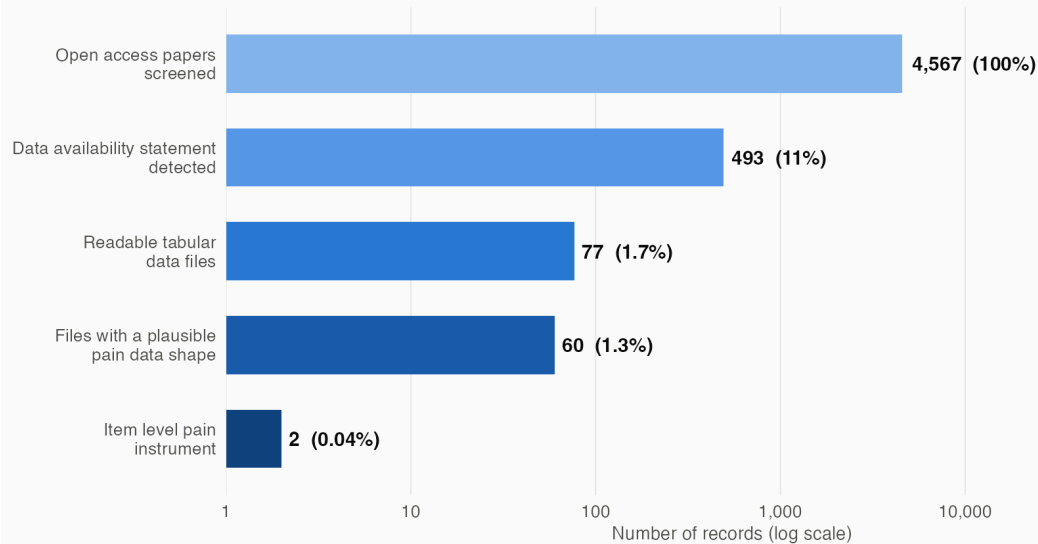
Supplementary bundles were pulled from the Europe PMC `supplementaryFiles` endpoint for every flagged paper, and repository links were resolved through the figshare, Zenodo, OSF and Dryad APIs. **102 tabular files** were retrieved, **77** of them machine readable, yielding **161 tables** once every sheet of every workbook was scored separately.

Screening scores tables on *shape* rather than column name regexes, because real shared files name the rating `PI`, `pain0` or `NRS` at least as often as `VAS`. A table scores highly when it holds a wide 0 to 100 column alongside a block of small integer

columns, which is what an item level pain dataset looks like. That produced a ranked review list of **60 tables**, which were then read by hand.

Pain data is shared. Item level pain data is not.

Europe PMC open access papers using a visual analogue scale for pain alongside a named multi item pain instrument. Availability statements detected with rtransparency. Log scale.

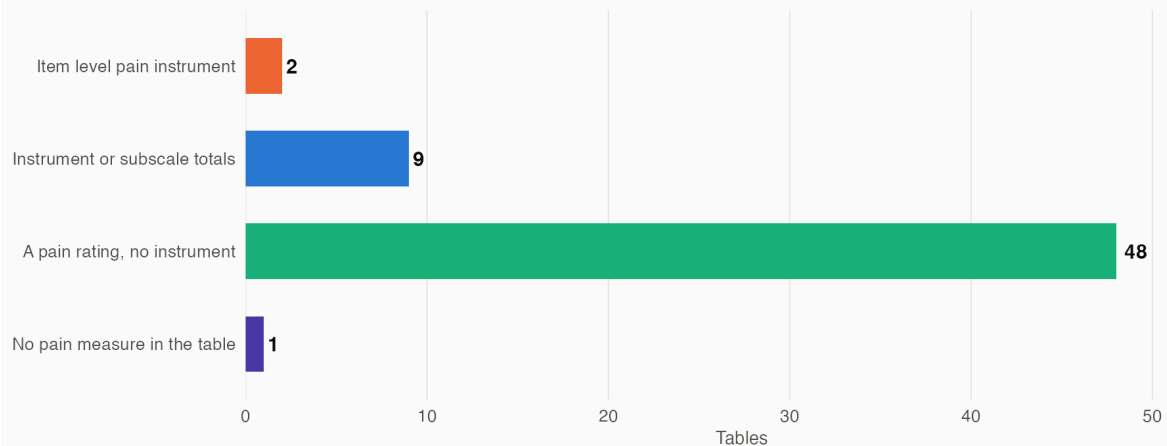


Two of 4,567 papers shared the item level responses an item response model needs. Everything else shared subscale or instrument totals, which cannot be modeled. Only 39 of the 493 availability statements named a repository or accession; 324 said the data are in the article.

WHAT THE 60 SHORTLISTED TABLES CONTAINED

What shared pain datasets actually contain

The 60 shortlisted tables, classified by the finest grain of pain measurement present. Only the top category can support an item response model.



Shortlist drawn from 102 tabular files retrieved from supplements and repositories, 77 of them readable.

The shape heuristic did its job as a ranker, but hand review was decisive. Most shortlisted tables earned their small integer columns from sex, ASA grade, comorbidity flags and treatment arm, not from questionnaire items. A targeted scan for the item block signatures of WOMAC, KOOS, SF-MPQ, BPI and PROMIS across all 161 tables returned **7 hits**, of which 5 were subscale totals.

Two datasets held genuine item level pain responses.

The two candidates, and why neither works

PMC5524330: KNEE OSTEOARTHRITIS, N = 386 OBSERVATIONS FROM 98 PATIENTS

Holds the full WOMAC as items: WOM A-1 to A-5 (pain), B-1 to B-2 (stiffness), C-1 to C-17 (function), plus a separate global VAS.

Inadmissible, on construct grounds. Each WOMAC item is itself scored on a 0 to 100 visual analogue scale. Building the latent metric out of VAS responses and then testing a VAS against it cannot detect a nonlinearity the two displays share, so the design is biased toward concluding that the scale is fine. The independent reference the method needs is exactly what this dataset lacks.

Two secondary problems would also have to be handled: 101 response categories per item need coarsening before a graded model will fit, and the 386 rows are four visits from 98 people.

PMC10695107: KNEE OSTEOARTHRITIS, N = 57

Holds the WOMAC pain subscale as five genuine 0 to 4 Likert items, measured at baseline and six weeks, with a separate VAS at each occasion. The items are independent of the VAS display, so this one is admissible in principle.

Underpowered and badly covered. Baseline VAS spans only 60 to 90; six week VAS spans 1 to 55. Neither occasion alone covers enough of the scale to fit a curve. Stacking both occasions covers 1 to 90 but leaves a hole between 55 and 60. The fitted

curve climbs from 8.3 to 24.5 as it crosses that empty stretch, peaking just beyond it, with a bootstrap band running from 12.9 to 32.2. That is the occasion gap, not scale geometry.

$r(\text{theta}, \text{VAS}) = 0.66$, which passes the construct check. Everything else fails.

The figure, in the two forms the evidence supports

Panel A is 12 simulated cohorts where the display compresses the top by construction. The estimator recovers the true curve: it reads about 13 near VAS 10, dips below 9 through the middle, and climbs back above 14 at the top. This is the validation that the pipeline measures what it claims to.

Panel B is PMC10695107. It is shown because the honest output of this stage is the yield, not a curve. The rise runs straight through a stretch of the scale where the dataset has no observations, and the band is wider than any effect worth reporting. Read it as a statement about the data supply, not about the VAS.

The figure appears in landscape on the following page.

Is the Visual Analogue Scale an Interval Scale?

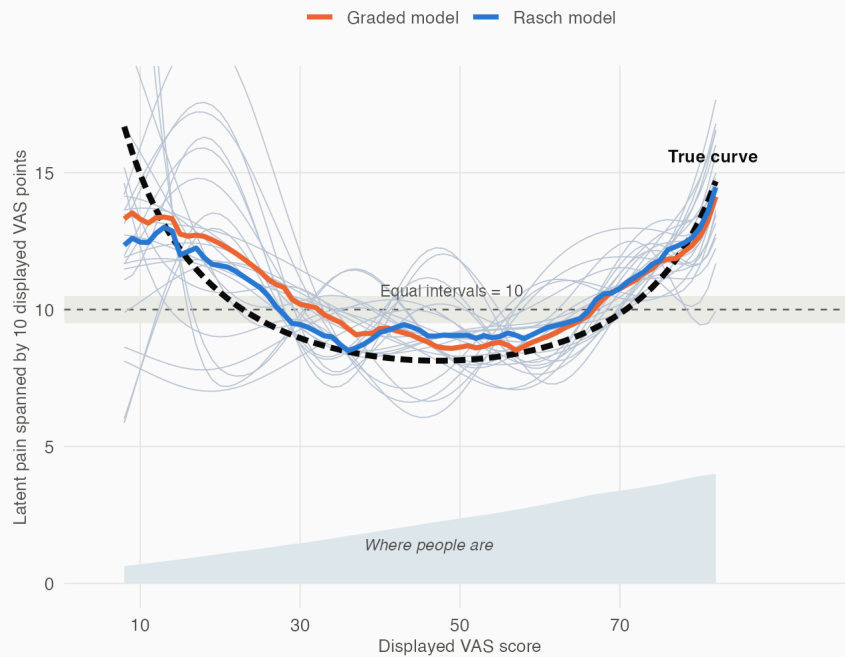
Is 10 points of pain always 10 points? The method works; the shared data cannot carry it.

Latent pain spanned by a 10 point interval on the 0 to 100 VAS, normalized so an equal interval scale reads 10.

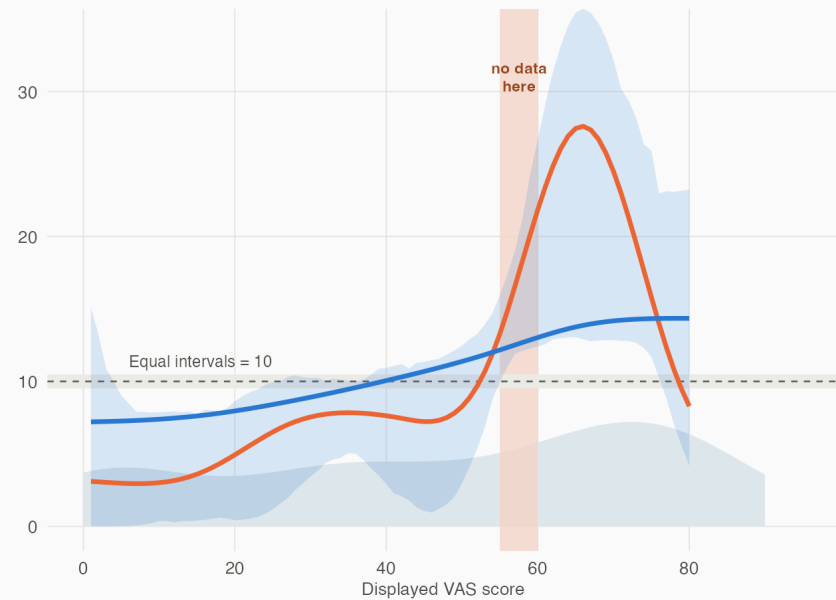
Left: 12 simulated cohorts where the display compresses the top by construction; the estimator recovers it.

Right: the single dataset out of 4,567 papers with item level pain responses. Its curve rises through a stretch of the scale where it has no data.

A. Simulated data, where the answer is known



B. The only qualifying dataset: n = 57, two occasions



Latent pain estimated by item response models fitted to a multi item pain instrument, excluding the VAS. Thin lines are cohorts, bold lines medians. Panel B band is a 150 replicate participant level bootstrap, 10th to 90th percentile. Note the panels use different y ranges: B reaches 34, A only 18. Panel B is not evidence about the VAS. It is evidence about the data supply.

Method

1. Fit a unidimensional IRT model to the multi item pain instrument, **excluding the VAS**. Both a graded response model and a Rasch model are fitted, so the figure can carry two bold lines the way the PISA plot does.
2. Take **WLE** factor scores as theta. EAP was rejected: it shrinks toward the prior mean, and hardest where test information is low, which is the tails. That would flatten the spline at both ends of the VAS and manufacture compression the scale does not have.
3. Regress theta on the displayed VAS with a monotone increasing spline (`scam`, `bs = "mpi"`).
4. $D(v) = f(v + 10) - f(v)$ is the latent distance spanned by a 10 point displayed interval starting at v . Under an interval scale D is flat.
5. Rescale so that the mean of D across the 5th to 95th percentile of the observed VAS distribution equals 10, so the curve reads directly as a ruler. Normalizing over the full 0 to 90 grid would let empty tails dominate.

Two contamination guards matter. Any column named as a visual analogue scale is barred from the item block, and so is any column correlating above 0.95 with the chosen display, which catches the same rating recorded twice in different units. BPI severity items are deliberately **not** barred despite being named `pain_worst` and `pain_now`: the graded model frees every item threshold, so a 0 to 10 item response is not assumed linear in latent pain, and excluding them would delete the best instrument in the corpus.

Repeated measures of one item are not items. A dataset whose proposed item block is mostly timepoint suffixed is flagged and refused unless explicitly overridden, or theta becomes average pain over the trial rather than pain at the moment the rating was given.

VALIDATION

`Rscript R/04_interval_curve.R` runs a self check: it simulates a display with known top end compression and asserts the estimator recovers an increasing interval curve with the normalization intact. It covers both the graded and the Rasch path.

What this means for the project

The exploratory version of this study cannot be done from journal supplements. Three routes remain, in order of cost:

1. **Cohorts that deposit item level data by design.** The Osteoarthritis Initiative and MOST both hold WOMAC at item level with pain ratings, at sample sizes three orders of magnitude above anything here. Both need a data use agreement rather than a download. This is the fastest route to the figure.

A direct search of Zenodo, Dryad and OSF by instrument name (`R/08_repo_search.R` , results in `out/repo_search_hits.csv`) surfaced five further human pain datasets worth opening by hand, including a Dryad deposit on staged bilateral knee arthroplasty pain and an OSF item content analysis of the Brief Pain Inventory interference subscale.

2. **Ask the two authors.** PMC10695107 has admissible items and only needs more participants; the same group may hold more.
3. **The psychophysics design.** Controlled thermal or pressure stimuli with pairwise comparisons, building the latent scale from a Bradley-Terry model rather than from another questionnaire. This is the most convincing version of the study and the most expensive.

There is also a second paper sitting in this output that needs no new data: of 4,567 open access pain papers, 10.8% share data and 0.04% share the item level responses required to re-analyze a psychometric question. Sharing rates are measured, rising, and precisely quantified here. That is a meta-research result in its own right.

Caveat carried from the design

The curve measures VAS intervals relative to the IRT logit metric. That metric is itself a modeling choice, the same assumption the PISA plot makes. It is the comparison that is informative, not the absolute units.

Reproducing

```
Rscript R/01_search.R          # Europe PMC, three strata
Rscript R/01b_search_plos.R    # journal policy stratum
Rscript R/01c_search_rest.R    # exhaust the broad stratum
Rscript R/02_flag.R            # XML download, rtransparency detection
Rscript R/02b_flag_extra.R
Rscript R/03_screen.R          # supplements, shape scoring, review list
Rscript R/03b_repos.R          # figshare, Zenodo, OSF, Dryad
Rscript R/08_repo_search.R     # direct repository search by instrument
Rscript R/04_interval_curve.R  # estimator plus self check
Rscript R/07_run.R             # curated analysis
Rscript R/09_figures.R         # figures 1 to 3
Rscript R/10_figure_main.R     # main figure
```

`data/` is gitignored. Downloaded datasets stay local and keep their original licenses and citations.